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3.3.7 Orthogonality of resulting Q

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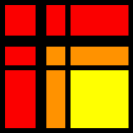
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No video for this unit.

Reading assignment

1/1 point (graded)



Read Unit 3.3.7 of the notes. [\[LINK\]](#)

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? [May need clarification on HW 3.3.7.1.](#)

[You indicate that students should write the MATLAB file `Assignments/Week03/matlab...](#)

3

Homework 3.3.7.1.

1/1 point (graded)

Previous programming assignments have the following routines for computing the QR factorization of a given matrix A :

- Classical Gram-Schmidt (CGS):

```
[ A_out, R_out ] = CGS_QR( A ) .
```

- Modified Gram-Schmidt (MGS):

```
[ A_out, R_out ] = MGS_QR( A ) .
```

- Householder QR factorization (HQR):

```
[ A_out, t_out ] = HQR( A ) .
```

- Form Q from Householder QR factorization:

```
Q = FormQ( A, t ) .
```

Use these to examine the orthogonality of the computed Q by writing the Matlab script (from scratch), in file

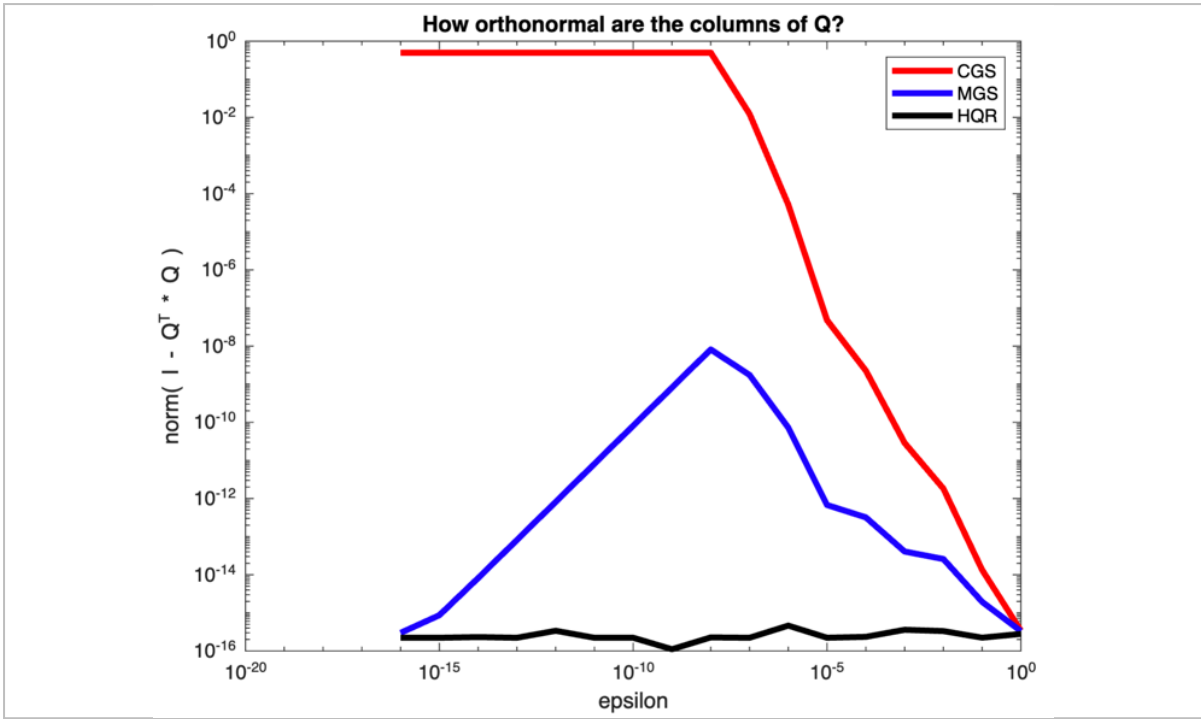
`Assignments/Week03/matlab/test_orthogonality.m` , for the matrix

$$\left(\begin{array}{c|c|c} 1 & 1 & 1 \\ \epsilon & 0 & 0 \\ 0 & \epsilon & 0 \\ 0 & 0 & \epsilon \end{array} \right) .$$

☒ done



Try [Week03/answers/test_orthogonality.m](#).



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Answers are displayed within the problem

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